

Dynamic modelling and input-energy comparison for the elevator system



Kun-Yung Chen^a, Ming-Shyan Huang^b, Rong-Fong Fung^{b,*}

^aInstitute of Engineering Science and Technology, National Kaohsiung First University of Science and Technology, 1 University Road, Yenchau, Kaohsiung 824, Taiwan

^bDepartment of Mechanical and Automation Engineering, National Kaohsiung First University of Science and Technology, 1 University Road, Yenchau, Kaohsiung 824, Taiwan

ARTICLE INFO

Article history:

Received 24 May 2012

Received in revised form 14 August 2013

Accepted 8 October 2013

Available online 26 October 2013

Keywords:

Dynamic modeling

Elevator system

Minimum energy

Trajectory planning

Sliding mode controller (SMC)

ABSTRACT

The elevator system driven by a permanent magnet synchronous motor (PMSM) is studied in this paper. The mathematical model of the elevator system includes the electrical and mechanical equations, and the dimensionless forms are derived for the purpose of practicable upward and downward movement. In this paper, the trapezoidal, cycloidal, five-degree (5-D) and seven-degree (7-D) polynomial and industry trajectories are designed and compared numerically in various motion and the absolute input energies. From numerical simulations, it is found that the trapezoidal trajectory consumes the minimum energy; the 7-D polynomial trajectory consumes the maximum one. The less end-point constraints are required, the less energy is consumed. Finally, the proposed sliding mode controller (SMC) is employed to demonstrate the robustness and well tracking control performance numerically.

© 2013 Elsevier Inc. All rights reserved.

1. Introduction

Nowadays, the elevator system has been a significant vertical transportation device for tall buildings, and has brought human convenience and efficiency in the recently decade years. However, the energy (electrical power, crude oil for example) has been consumed and exploited greatly for many years. The energy consumption of the elevator system is the most part in the power consumption of tall buildings. Therefore, the energy saving of the elevator system is an interesting study and meaningful topic. In the previous studies, the authors only emphasized that the string vibration suppression control [1,2], but the topic in energy consumption was not discussed. The robust control algorithms [3–6] also were employed for the elevator system, where the car trajectory was tracked by the robust controller, and the numerical simulations and experimental results demonstrated that the proposed controller was superior. In the previous papers [1–6], what kind of trajectories can save energy was seldom discussed. Furthermore, the end-point constraints of trajectories were not studied for the comparisons in the input electrical energies.

Schlemmer and Agrawal [7] proposed the elevator to transport passengers in a near minimum time while satisfying elevator's intrinsic dynamic constraints, such as allowable hoist torque/power, and extrinsic comfort constraints, such as allowable acceleration and deceleration. Mutoh et al. [8] proposed a induction motor driving controller for the drive system of elevators on the basis of simulations and experiments to improve performance of elevators. These papers [7,8] neither define the energy nor to propose energy-saving controllers for the minimum energy/time trajectories. Moreover, the previous

* Corresponding author. Tel.: +886 936659758; fax: +886 7 6011066.

E-mail address: rffung@nkfust.edu.tw (R.-F. Fung).

researches [9–11] even proposed the minimum-energy control method to the energy-saving problems, but the energy definition is not the physical one.

The most definition of cost function in an optimal control problem is the sum of square control effort of the system, but not the physical energy. Fortunately, Kokotovic and Singh [12] presented the minimum-energy control for a nonlinear second-order model of a ground transportation vehicle with dc traction motor. They proposed the electrical energy as the cost function for the dc motor drive system. Huang et al. [13] proposed a minimum-energy point-to-point (PTP) trajectory planning method for a motor-toggle servomechanism, and defined the electrical energy as the fitness function for the real-coded genetic algorithm method. In [13], a nine-degree polynomial trajectory and the initial- and final-state constraints were discussed. Zhua and Teppo [14] proposed a novel scaled model to simulate the linear lateral dynamics of a hoist cable with variable length in a high-rise, high-speed elevator, where the position function is given by a fifth order polynomial and is divided into seven regions, and the prototype movement profile was described clearly and compared with other trajectories.

The trajectory planning topics also were discussed in the previous studies [15–19]. The authors emphasized the optimal time-jerk trajectory planning to the robot manipulators. In order to obtain the optimal trajectory, two objective functions composed of two terms are minimized. The high degree polynomial trajectory was planned and implemented in the manipulator [20]. The linear acceleration profiles of end-effector were planned for the polynomials of degree 9, 7 and 5. In these studies [15–20], the energy consumptions of the trajectories for the robot manipulator were not considered, and the initial- and final-time constraints were also not discussed and compared.

In this paper, we propose the complete mechatronic model with the electrical equation of a PMSM and the mechanical equation of an elevator system. The car's displacement and travel time of the elevator system is transferred to the dimensionless form for the upward and downward movement. The energy definition of the elevator system is proposed and the total input energies are compared among various trajectories, which include the trapezoidal, cycloidal, 5-D, 7-D polynomial and industry trajectories. The end-point constraints are not the same for various trajectories, and then their input energies are different. This paper is going to find the relationship between the initial- and final-time constraints and the total absolute input energies from numerical simulations. Finally, the proposed SMC demonstrates that the controller has the robustness and well tracking control performance numerically.

2. Modeling of the elevator system

Firstly, the electrical equation of the PMSM is given, and then the mechatronic elevator system driven by the PMSM is formulated. The string mass of the elevator system is also considered as the variable external force acting on the PMSM. The dimensionless process is applied and the elevator model becomes a normalized form. The energy definition of the PMSM is also defined.

2.1. Model of the PMSM

The stator flux-linkage equation of a PMSM [21] can be described as follows:

$$v_q = R_s i_q + L_q di_q/dt + \lambda_d \omega_r, \quad (1a)$$

$$v_d = R_s i_d + L_d di_d/dt - \lambda_q \omega_r, \quad (1b)$$

where v_d and v_q are the stator voltages, i_d and i_q are the stator currents, L_d and L_q are the inductances, λ_d and λ_q are the stator flux linkages of the d and q axis, respectively. R_s is the stator resistance, d/dt is total differentiation with respect to time, and ω_r is the rotor angular speed.

The electromagnetic torque τ_e can be described as:

$$\tau_e = T_r + B_m \omega_r + J_m \dot{\omega}_r. \quad (2)$$

The applied torque can be obtained as follows:

$$T_r = K_t i_q - J_m \dot{\omega}_r - B_m \omega_r. \quad (3)$$

Fig. 1 shows the PMSM including a gear speed-reducer. n is the gear ratio number and can be described as follows:

$$n = n_a/n_b = \omega_r/\omega = T_l/T_r, \quad (4)$$

where n_a and n_b are the gear number, T_l is the torque applied on the sheave, and ω is the angular speed of the sheave. Then, the electrical torque and the applied torque are given respectively as:

$$T_r = K_t i_q - J_m \dot{\omega}_r - B_m \omega_r = K_t i_q - J_m n \dot{\omega} - B_m n \omega, \quad (5)$$

$$T_l = n T_r = n K_t i_q - J_m n^2 \dot{\omega} - B_m n^2 \omega. \quad (6)$$

2.2. Elevator system driven by a PMSM

In Fig. 2, the elevator system is shown and the sheave rotation is driven by a PMSM. The main string passes over the drive sheave and is attached to a counterweight. The purpose of the counterweight is to compensate the elevator's weight and it

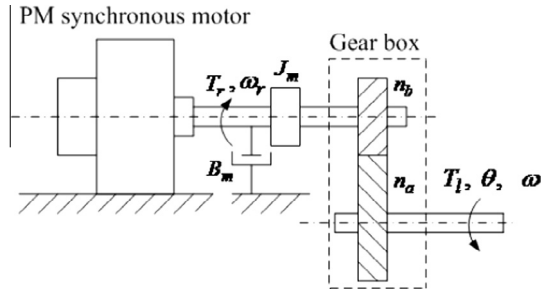


Fig. 1. The gear speed-reducer.

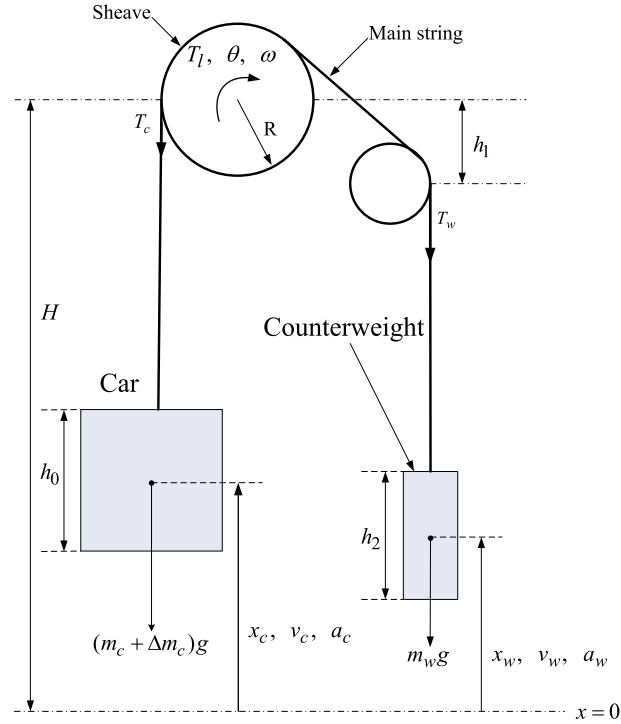


Fig. 2. The physical model of the elevator system.

can reduce the torque on the motor when the elevator is stationary or moving under constant speed. In Fig. 2, T_l is the torque applied on the drive sheave, θ is the angular displacement of the sheave, ω is the angular speed of the sheave, R is the radius of the sheave, H is the length between the original position and sheave center point. T_c and T_w are the tensions in the car and counterweight, respectively. The mass per unit length of the string is ρ (kg/m).

In this paper, we neglect the sag effect and the cable is considered as rigid body. Therefore, the car displacement, speed and acceleration with respect to the sheave angular displacement, angular speed and angular acceleration can be respectively described as:

$$x_c = x_{c0} + \theta R, \quad v_c = \dot{\theta}R = \omega R, \quad a_c = \ddot{\theta}R = \alpha R, \quad (7a-c)$$

where x_{c0} is the car's initial position, and α is the angular acceleration of the drive sheave.

Similarly, the displacement, speed and acceleration of the counterweight can be respectively described as:

$$x_w = x_{w0} - \theta R, \quad v_w = -\dot{\theta}R = -\omega R, \quad a_w = -\ddot{\theta}R = -\alpha R, \quad (8a-c)$$

where x_{w0} is the initial position of the counterweight.

The tension T_c on the car side can be found as:

$$T_c = (m_c + \Delta m_c)g + (H - h_0 - x_c)\rho g + c v_c + [m_c + \Delta m_c + (H - h_0 - x_c)\rho]a_c. \quad (9a)$$

Eqs. (7a)–(7c) are substituted into Eq. (9a) and we can obtain:

$$T_c = m_c + \Delta m_c g + (H - h_0 - x_{c0} - \theta R)\rho g + c\omega R + [m_c + \Delta m_c + (H - h_0 - x_{c0})\rho - \theta R\rho]\alpha R, \quad (9b)$$

where m_c is the car mass, Δm_c is the passenger mass and c is the viscous coefficient of the car's guide. The tension T_w on the counterweight can be found as:

$$T_w = m_w g + (H - h_1 - h_2 - x_w)\rho g + [m_w + (H - h_1 - h_2 - x_w)\rho]a_w. \quad (10a)$$

Equations (8a~c) are substituted into Eq. (10a) and we can obtain:

$$T_w = m_w g + (H - h_1 - h_2 - x_{w0} + \theta R)\rho g + [m_w + (H - h_1 - h_2 - x_{w0})\rho + \theta R\rho](-\alpha R), \quad (10b)$$

where m_w is the mass of the counterweight.

According to the dynamic equation in rotation, we can obtain the follow balance equation as:

$$T_l + T_w R - T_c R = J\alpha, \quad (11)$$

where J is the moment of inertia of the drive sheave. Equations (6), (9b) and (10b) are substituted into Eq. (11), and we can obtain the dynamic equation of the elevator system driven by a PMSM as:

$$\begin{aligned} & \{J + n^2 J_m + (m_c + \Delta m_c)R^2 + m_w R^2 + [2H - (h_0 + h_1 + h_2) - (x_{c0} + x_{w0})]\rho R^2\}\alpha \\ & = 2\rho g R^2 \theta - (n^2 B_m + cR^2)\omega + nK_t i_q + [(h_0 + x_{c0} - h_1 - h_2 - x_{w0})\rho + (m_w - m_c)]gR - \Delta m_c gR. \end{aligned} \quad (12)$$

Here, the total moment of inertia J_t and the external torque $\tau_d(t)$ are set respectively as:

$$J_t = J + n^2 J_m + (m_c + \Delta m_c)R^2 + m_w R^2 + [2H - (h_0 + h_1 + h_2) - (x_{c0} + x_{w0})]\rho R^2, \quad (13a)$$

$$\tau_d(t) = [(h_0 + x_{c0} - h_1 - h_2 - x_{w0})\rho + (m_w - m_c)]gR - \Delta m_c gR, \quad (13b)$$

and Eq. (12) can be rewritten as:

$$J_t \alpha = 2\rho g R^2 \theta - (n^2 B_m + cR^2)\omega + nK_t i_q + \tau_d(t). \quad (14)$$

According to Eqs. (5) and (1), we have:

$$v_q = i_q R_s + L_q (di_q/dt) + \lambda_d n \omega. \quad (15)$$

From Eqs. (14) and (15), we can have the state space matrix form as:

$$\dot{\mathbf{X}} = \mathbf{A}\mathbf{X} + \mathbf{B}u + \mathbf{D}, \quad (16)$$

where $\mathbf{X} = [\theta \quad \omega \quad i_q]^T$, $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 2\rho g R^2/J_t & -(n^2 B_m + cR^2)/J_t & nK_t/J_t \\ 0 & -\lambda_d n/L_q & -R_s/L_q \end{bmatrix}$, $\mathbf{B} = [0 \quad 0 \quad 1/L_q]^T$, $\mathbf{D} = [0 \quad \tau_d(t)/J_t \quad 0]^T$ and $u = v_q$.

2.3. The dimensionless governing equations

For convenience in determining the influence of parameters of the mechatronic system, we define the relationship between translation displacement h of the car and angular displacement θ_m of the drive sheave as:

$$h = \theta_m R = l|n_j - n_i| > 0, \quad (17)$$

where θ_m is the angular displacement, l is the high of each floor, n_i and n_j are the numbers of departure and arrival floors, respectively. It is assumed that T is the travel time from the departure floor to the arrival floor, and the car's average speed can be described as:

$$v_a = \omega_a R, \quad (18a)$$

where ω_a is the average angular speed of the drive sheave.

The dimensionless forms [22] of the angular displacement and time are respectively defined as follows:

$$\Theta = \theta/\theta_m \quad \text{and} \quad \tau = \omega_a t/\theta_m. \quad (18b-c)$$

Therefore, we can find the dimensionless forms of angular speed, acceleration and jerk as:

$$\dot{\Theta} = \omega_a \Theta_\tau, \quad \ddot{\Theta} = \omega_a^2 \Theta_{\tau\tau}/\theta_m \quad \text{and} \quad \dddot{\Theta} = \omega_a^3 \Theta_{\tau\tau\tau}/\theta_m^2. \quad (19a-c)$$

Then, the dimensionless parameters and variables are given as:

$$\begin{aligned} \bar{J} &= J_t \omega_a^2 / (nK_t i_m \theta_m), \quad \bar{a} = 2\rho g R^2 \theta_m / (nK_t i_m), \quad \bar{b} = (n^2 B_m + cR^2) \omega_a / (nK_t i_m), \quad \bar{I} = i_q / i_m, \quad \bar{d} = \bar{d}_0 - \Delta \bar{d}, \quad \bar{d}_0 \\ &= [(h_0 + x_{c0} - h_1 - h_2 - x_{w0})\rho + (m_w - m_c)]gR / (nK_t i_m), \quad \Delta \bar{d} = \Delta m_c gR / (nK_t i_m). \end{aligned} \quad (19d-j)$$

where $i_m > 0$ is the maximum input current. The mechanical equation (14) can be written with the dimensionless form as:

$$\bar{J}\Theta_{\tau\tau} = \bar{a}\Theta - \bar{b}\Theta_{\tau} + \bar{I} + \bar{d}. \quad (20)$$

From the Eq. (19g), we can find that:

$$i_q = \bar{I}_m, \quad di_q/dt = (\omega a i_m / \theta_m) d\bar{I} / d\tau. \quad (21a-b)$$

Let $v_m > 0$ is the maximum input voltage. Equations (19a), (21a-b) are substituted into Eq. (15), and we can obtain the dimensionless electrical equation as:

$$\bar{U} = \bar{R}_s \bar{I} + \bar{L}_q \bar{I}_{\tau} + \bar{\lambda}_d \Theta_{\tau}, \quad (22)$$

where $\bar{U} = v_q / v_m$, $\bar{R}_s = R_s i_m / v_m$, $\bar{L}_q = L_q \omega a i_m / (v_m \theta_m)$, $\bar{\lambda}_d = n \lambda_d \omega a / v_m$.

Finally, we can find the dimensionless state space matrix form of the elevator system driven by the PMSM from Eqs. (20) and (22) as:

$$\bar{\mathbf{X}}_{\tau} = \bar{\mathbf{A}}\bar{\mathbf{X}} + \bar{\mathbf{B}}\bar{\mathbf{U}} + \bar{\mathbf{D}}, \quad (23)$$

where $\bar{\mathbf{X}} = [\Theta \quad \Theta_{\tau} \quad I]^T$, $\bar{\mathbf{A}} = \begin{bmatrix} 0 & 1 & 0 \\ \bar{a}/\bar{J} & -\bar{b}/\bar{J} & 1/\bar{J} \\ 0 & -\bar{\lambda}_d/\bar{L}_q & -\bar{R}_s/\bar{L}_q \end{bmatrix}$, $\bar{\mathbf{B}} = [0 \quad 0 \quad 1/\bar{L}_q]^T$, and $\bar{\mathbf{D}} = [0 \quad \bar{d}/\bar{J} \quad 0]^T$.

2.4. The input energy

The energy transferring between electrical and mechanical systems has significant meaning in this paper. For the energy-saving goal, it is important to find the energy equations for the elevator system. The mechanical equation (14) can be rewritten as:

$$i_q = J_t \alpha / n K_t + (n^2 B_m + c R^2) \omega / n K_t - 2 \rho g R^2 \theta / n K_t - \tau_d(t) / n K_t. \quad (24)$$

Substituting Equation (24) into Eq. (15) and integrating from 0 to T , we obtain the energy equation of the elevator system:

$$E_i = E_d + E_p + E_o, \quad (25)$$

where $E_i = \int_0^T v_q i_q dt$ is the input energy,

$E_d = \int_0^T \{R_q^2 + (\lambda_d / K_t)(c R^2 + n^2 B_m) \omega^2 - (\lambda_d / K_t) \tau_d(t) \omega\} dt$ is the dissipation energy,

$E_p = \int_0^T [L_q (di_q / dt) i_q - (\lambda_d / K_t) 2 \rho g R^2 \theta \omega] dt$ is the potential energy and

$E_o = \int_0^T (\lambda_d / K_t) (J_t) \omega \dot{\omega} dt$ is the output energy.

The absolute input energy can be described as:

$$|E_i| = \int_0^T |v_q i_q| dt, \quad (26a)$$

where

$$i_q = [J_t \alpha + (n^2 B_m + c R^2) \omega - 2 \rho g R^2 \theta - \tau_d(t)] / n K_t, \quad (26b)$$

$$v_q = L_q J_t \dot{\alpha} / n K_t + [R_s J_t + L_q (n^2 B_m + c R^2)] \alpha / n K_t + (R_s (n^2 B_m + c R^2) / n K_t + \lambda_d n) \omega - R_s \tau_d(t) / n K_t - L_q [d\tau_d(t) / dt] / n K_t. \quad (26c)$$

If the parameters J_t , B_m , R , K_t , L_q , R_s and λ_d are all known, the electrical state i_q and v_q can be obtained by the output responses θ , ω , α and $\tau_d(t)$, and then we can find the absolute input energy by the system responses and external torques. Finally, the input energy can be written in the normalized form as:

$$E_i = \int_0^T i_q v_q dt = [(i_m v_m \theta_m) / \omega_a] \int_0^1 (\bar{I} \bar{U}) d\tau. \quad (27)$$

3. Point-to-point trajectory planning

In this section, we choose three basic trajectories including trapezoidal, cycloidal, and polynomial trajectories, and to compare with the industry prototype elevator trajectory [14] for the energy consumption.

3.1. Trapezoidal trajectory

A very common method to obtain a trajectory with continuous speed profile is to use linear motions with parabolic blends, and the trajectory can be characterized by the typical trapezoidal speed profile. For the trapezoidal trajectory, the car's initial and final conditions of displacements and speeds at the start and final times are given as:

$$x_c(0) = 0, x_c(T) = h, \quad v_c(0) = 0 \quad \text{and} \quad v_c(T) = 0. \quad (28a-d)$$

The initial and final conditions of the sheave angular displacement and speeds at the start and final times are given as:

$$\theta(0) = x_c(0)/R, \quad \theta(T) = h/R, \quad \omega(0) = v_c(0)/R \quad \text{and} \quad \omega(T) = v_c(T)/R, \quad (29a-d)$$

where time T is the final time and h is the car's total displacement.

The final time T and the total displacement h of the car are variables from one floor to another floor. Therefore, the normalized forms of Θ , Θ_τ and $\Theta_{\tau\tau}$ are employed, the dimensionless trapezoidal trajectory are shown in Fig 3. Firstly, the normalized form of the Θ is 1 and we can find the dimensionless variables as:

$$1 = \Theta = \tau_1 \Theta_\tau^0 + (\tau_2 - \tau_1) \Theta_\tau^0, \quad \Theta_\tau^0 = 1/\tau_2, \quad \alpha_n = 1/(\tau_1 \tau_2), \quad \tau_1 = 0.2 \quad \text{and} \quad \tau_2 = 0.8, \quad (30a-d)$$

where Θ_τ^0 is the constant dimensionless angular speed, α_n is the dimensionless acceleration.

There are three phases including acceleration phase, constant speed phase and deceleration phase for the trapezoidal trajectory, and each dimensionless phase can be formulated as follows.

3.1.1. Acceleration phase

For the acceleration phase, $\tau \in [0, \tau_1]$ and τ_1 is the end time of the acceleration phase. The sheave dimensionless angular displacement, speed and acceleration are respectively expressed as:

$$\Theta(\tau) = (1/2) \Theta_{\tau\tau} \tau^2 \text{sign}(n_j - n_i), \quad \Theta_\tau(\tau) = \Theta_{\tau\tau} \tau \text{sign}(n_j - n_i), \quad \Theta_{\tau\tau}(\tau) = \alpha_n \text{sign}(n_j - n_i). \quad (31a-c)$$

where $\text{sign}(n_j - n_i)$ is the sign function of the sheave rotating direction.

3.1.2. Constant speed phase

For the constant speed phase, $\tau \in [\tau_1, \tau_2]$. The sheave dimensionless angular displacement, speed and acceleration are respectively expressed as:

$$\Theta(\tau) = \Theta(\tau_1) + \Theta_\tau^0 (\tau - \tau_1) \text{sign}(n_j - n_i), \quad \Theta_\tau(\tau) = \Theta_\tau^0 \text{sign}(n_j - n_i), \quad \Theta_{\tau\tau}(\tau) = 0. \quad (32a-c)$$

3.1.3. Deceleration phase

For the deceleration phase, $\tau \in [\tau_2, 1]$. The sheave dimensionless angular displacement, speed and acceleration are respectively expressed as:

$$\begin{aligned} \Theta(\tau) &= \Theta(\tau_2) + \Theta_\tau^0 (\tau - \tau_2) \text{sign}(n_j - n_i) - \frac{1}{2} \alpha_n (\tau - \tau_2)^2 \text{sign}(n_j - n_i), \quad \Theta_\tau(\tau) \\ &= \Theta_\tau^0 \text{sign}(n_j - n_i) - \alpha_n (\tau - \tau_2) \text{sign}(n_j - n_i), \quad \Theta_{\tau\tau}(\tau) = -\alpha_n \text{sign}(n_j - n_i). \end{aligned} \quad (33a-c)$$

3.2. Cycloidal trajectory

A continuous acceleration profile can be obtained by the cycloidal trajectory [23]. The normalized forms of angular displacement, speed, acceleration and jerk of the cycloidal trajectory are rewritten as:

$$\begin{aligned} \Theta(\tau) &= (\tau - (1/2\pi) \sin 2\pi\tau) \text{sign}(n_j - n_i), \quad \Theta_\tau(\tau) = (1 - \cos 2\pi\tau) \text{sign}(n_j - n_i), \quad \Theta_{\tau\tau}(\tau) \\ &= [2\pi \sin 2(\pi\tau)] \text{sign}(n_j - n_i) \quad \text{and} \quad \Theta_{\tau\tau\tau}(\tau) = [4\pi^2 \cos(2\pi\tau)] \text{sign}(n_j - n_i). \end{aligned} \quad (34a-d)$$

Based on the elevator system, the sheave angular displacement, speed, acceleration and jerk of the cycloidal motion are respectively expressed as:

$$\theta(t) = \theta_m \Theta(\tau), \quad \omega(t) = \theta_m \Theta_\tau(\tau)/T, \quad \alpha(t) = \theta_m \Theta_{\tau\tau}(\tau)/T^2, \quad j(t) = \theta_m \Theta_{\tau\tau\tau}(\tau)/T^3, \quad (34e-h)$$

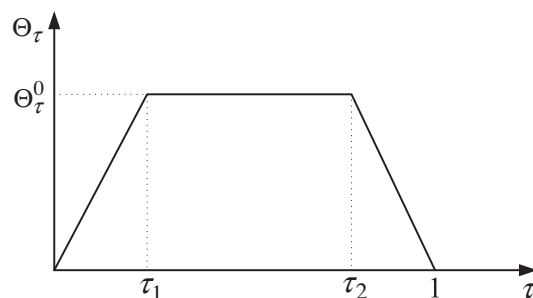


Fig. 3. The dimensionless speed of the trapezoidal trajectory.

where $\tau = t/T$.

3.3. Polynomial trajectory

A trajectory planning problem can be easily solved by considering a polynomial function for the trajectory planning with constraint conditions at the initial and final times. The normalized forms of the sheave angular displacement, speed, acceleration and jerk can be described as:

$$\Theta(\tau) = \text{sign}(n_j - n_i)[c_0 + c_1\tau + c_2\tau^2 + c_3\tau^3 + c_4\tau^4 + \cdots + c_n\tau^n], \quad (35a)$$

$$\Theta_\tau(\tau) = \text{sign}(n_j - n_i)[c_1 + 2c_2\tau + 3c_3\tau^2 + 4c_4\tau^3 + \cdots + nc_n\tau^{n-1}], \quad (35b)$$

$$\Theta_{\tau\tau}(\tau) = \text{sign}(n_j - n_i)[2c_2 + 6c_3\tau + 12c_4\tau^2 + \cdots + n(n-1)c_n\tau^{n-2}], \quad (35c)$$

$$\Theta_{\tau\tau\tau}(\tau) = \text{sign}(n_j - n_i)[6c_3 + 24c_4\tau + \cdots + n(n-1)(n-2)c_n\tau^{n-3}], \quad (35d)$$

where $0 \leq \tau \leq 1$. If the angular displacements, speeds, accelerations are constrained at the initial and final times. There are six constraint conditions and the 5-D polynomial is chosen. The six coefficients (c_0, c_1, c_2, c_3, c_4 and c_5) can be solved exactly by the six constraint conditions.

The six constraint conditions are expressed as:

$$\Theta(0) = 0, \quad \Theta(1) = 1, \quad \Theta_\tau(0) = v_0, \quad \Theta_\tau(1) = v_1, \quad \Theta_{\tau\tau}(0) = a_0 \quad \text{and} \quad \Theta_{\tau\tau}(1) = a_1. \quad (36a-f)$$

By using (36a-f), we can find the coefficients of the 5-D polynomial as:

$$\begin{aligned} c_0 &= x_0, \quad c_1 = v_0, \quad c_2 = a_0/2, \quad c_3 = [20 - (8v_1 + 12v_0) - (3a_0 - a_1)]/2, \quad c_4 \\ &= [-30 + (14v_1 + 16v_0) + (3a_0 - 2a_1)]/2, \quad c_5 = [12 - 6(v_1 + v_0) + (a_1 - a_0)]/2. \end{aligned}$$

In addition, there are eight constraint conditions including that displacements, speeds, accelerations and jerks are fixed at the initial and final times. Therefore, a 7-D polynomial is chosen and eight coefficients can be solved exactly by the eight constraint conditions. The eight constraints include the above six constraints Eqs. (36a-f), the other two constraints are expressed as:

$$\Theta_{\tau\tau\tau}(0) = j_0 \quad \text{and} \quad \Theta_{\tau\tau\tau}(1) = j_1. \quad (37a-b)$$

By using Eqs. (36a-f) and (37a-b), we can find the coefficients of the 7-D polynomial as:

$$\begin{aligned} c_0 &= x_0, \quad c_1 = v_0, \quad c_2 = a_0/2, \quad c_3 = j_0/6, \quad c_4 = \{210 - [(30a_0 - 15a_1) + (4j_0 + j_1) + 120v_0 + 90v_1]\}/6, \quad c_5 \\ &= \{-168 + [(20a_0 - 14a_1) + (2j_0 + j_1) + 90v_0 + 78v_1]\}/5, \quad c_6 \\ &= \{420 - [(45a_0 - 39a_1) + (4j_0 + 3j_1) + 216v_0 + 204v_1]\}/6, \quad c_7 \\ &= \{-120 + [(12a_0 - 12a_1) + (j_0 + j_1) + 60v_0 + 60v_1]\}/6. \end{aligned}$$

By using the geometric and time-scaling relations, we can transfer the dimensionless form to the real system as:

$$\theta(t) = \theta_m \Theta(\tau), \quad \dot{\theta}(t) = \omega_a \Theta_\tau(\tau), \quad \ddot{\theta}(t) = \omega_a^2 \Theta_{\tau\tau}(\tau)/\theta_m, \quad \dddot{\theta}(t) = \omega_a^3 \Theta_{\tau\tau\tau}(\tau)/\theta_m^2. \quad (38a-d)$$

$$x_c(t) = x_{c0} + R\theta_m \Theta(\tau), \quad v_c(t) = R\omega_a \Theta_\tau(\tau), \quad a_c(t) = R\omega_a^2 \Theta_{\tau\tau}(\tau)/\theta_m, \quad j_c(t) = R\omega_a^3 \Theta_{\tau\tau\tau}(\tau)/\theta_m^2. \quad (39a-d)$$

3.4. Industry trajectory

In order to compare various trajectories, an industry profile [14] is also referred in this paper, and its length of the elevator movement is $h = 150$ m. It is assumed that the car's maximum speed is 5 m/s, and the maximum acceleration is 0.75 m/s^2 . The car's initial displacement, speed, acceleration and jerk are zero, the final displacement, speed, acceleration and jerk are $x_c(T) = 150$ m, $v_c(T) = 0$ m/s, $a_c(T) = 0 \text{ m/s}^2$ and $j_c(T) = 0 \text{ m/s}^3$. The position function of the elevator is divided into seven regions, and its detail equations of the displacement, speed, acceleration and jerk can refer to Zhua and Teppo [14].

4. Design of SMC

Rewriting the dimensionless, single-input–single output (SISO) elevator system (23) as:

$$\ddot{\bar{\mathbf{X}}}_\tau = \bar{\mathbf{A}}\bar{\mathbf{X}} + \bar{\mathbf{B}}\bar{\mathbf{U}} + \bar{\mathbf{D}}_0 - \Delta\bar{\mathbf{D}}. \quad (40)$$

where $\bar{\mathbf{D}}_0 = [0 \quad \bar{d}_0/\bar{J} \quad 0]^T$ and $\Delta\bar{\mathbf{D}} = [0 \quad \Delta\bar{d}/\bar{J} \quad 0]^T$.

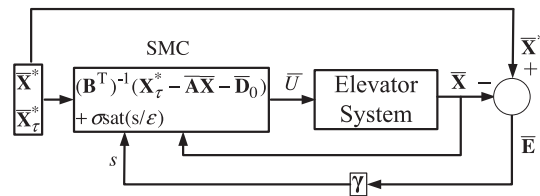


Fig. 4. The block diagram of the dimensionless SMC system.

$\bar{U}$ is the dimensionless control input effort. It is assumed that the matrices $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$ and $\bar{\mathbf{D}}_0$ are exactly known and the disturbance $\Delta\bar{\mathbf{D}}$ is unknown. The control problem is to find a control law so that the state $\bar{\mathbf{X}}$ can track the desired trajectories $\bar{\mathbf{X}}^*$ in the presence of the uncertainties. Let the tracking error vector be:

$$\bar{\mathbf{E}} = \bar{\mathbf{X}}^* - \bar{\mathbf{X}}, \quad (41)$$

where $\bar{\mathbf{X}}^* = [\Theta^* \quad \Theta_\tau^* \quad \bar{I}^*]^T$. Now, a dimensionless sliding surface function is defined in the dimensionless state space by the scalar function $s(\bar{\mathbf{X}}; \tau)$, where:

$$s(\bar{\mathbf{X}}; \tau) = \gamma \bar{\mathbf{E}}, \quad (42a)$$

where $\gamma = [\gamma_1 \quad \gamma_2 \quad 1]$, $\gamma_1 > 0$ and $\gamma_2 > 0$.

Differentiating $s(\bar{\mathbf{X}}; \tau)$ with respect to τ , we have:

$$s_\tau = \gamma(\dot{\bar{\mathbf{X}}}^* - \bar{\mathbf{A}}\bar{\mathbf{X}} - \bar{\mathbf{B}}\bar{U} - \bar{\mathbf{D}}_0 + \Delta\bar{\mathbf{D}}). \quad (42b)$$

The tracking problem is to find a dimensionless control law $\bar{U}$ so that the state $\bar{\mathbf{X}}$ remaining on the surface $s(\bar{\mathbf{X}}; \tau) = 0$ for all $\tau \geq \tau_0$, where τ_0 is the dimensionless time when the sliding motion occurs. In design of the SMC system, the equivalent control law $\bar{U}_{eq}$, which will determine the system dynamics on the sliding surface, should be found first. Then the equivalent control law is derived by recognizing:

$$s_\tau|_{\bar{U}=\bar{U}_{eq}} = 0. \quad (43)$$

Substituting (42b) into (43), and assuming the uncertainty $\Delta\bar{\mathbf{D}}$ are zero, then:

$$\gamma(\dot{\bar{\mathbf{X}}}_\tau^* - \bar{\mathbf{A}}\bar{\mathbf{X}} - \bar{\mathbf{B}}\bar{U}_{eq} - \bar{\mathbf{D}}_0) = 0. \quad (44a)$$

The equivalent control input can be found as:

$$\bar{U}_{eq} = (\bar{\mathbf{B}}^T)^{-1}(\dot{\bar{\mathbf{X}}}_\tau^* - \bar{\mathbf{A}}\bar{\mathbf{X}} - \bar{\mathbf{D}}_0). \quad (44b)$$

Based on the developed sliding surface, a switching control law which satisfies the hitting condition and guarantees the existence of the sliding mode is designed. Now, the sliding mode controller is proposed in the following:

$$\bar{U} = \bar{U}_{eq} + \bar{U}_n, \quad (45)$$

where $\bar{U}_n = \sigma \text{sat}(s/\varepsilon)$, $\sigma, \varepsilon > 0$, and $\text{sat}(s/\varepsilon) = \begin{cases} 1; s \geq \varepsilon, \\ s/\varepsilon; -\varepsilon < s < \varepsilon, \\ -1; s \leq -\varepsilon. \end{cases}$

Finally, the block diagram of the dimensionless sliding mode controller with an elevator system is shown in Fig. 4. The dimensionless control input $\bar{U}$ and variables $\bar{\mathbf{X}}$ can be transferred as dimension variables by Eq. (19) and (22), respectively.

5. Numerical simulation

In the numerical simulation, the parameters of an elevator drive system are selected as follows: $J_m = 8 \times 10^{-4}$ (kgm²), $B_m = 10^{-3}$ (N ms/rad), $K_t = 2$ (Nm/A), $R_s = 5$ (Ω), $\lambda_d = 0.05$ (N - s/rad), $L_q = 10^{-2}$ (H), $J = 0.1$ (kg m²), $R = 0.25$ (m), $m_c = 60$ (kg), $m_w = 50$ (kg), $c = 0.5$ (Ns/m), $n = 1$, $H = 200$ (m), $h_0 = 2$ (m), $h_1 = 1$ (m), $h_2 = 1$ (m), $l = 4$ (m), $\rho = 0.1$ (kg/m), $i_m = 100$ (A), and $u_m = 400$ (V). The dimensionless parameters are given as $\bar{J} = 0.0191$, $\bar{B} = 6.45 \times 10^{-4}$, $\bar{R}_s = 1.25$, $\bar{L}_q = 3.125 \times 10^{-4}$, and $\bar{\lambda}_d = 5 \times 10^{-4}$.

5.1. Comparisons of transient responses and input-energies

In the numerical simulations, we choose the departure floor $n_i = 1$, arrival floor $n_j = 3$ and the car average speed $v_a = 1$ (m/s). The dimensionless transient responses of the trapezoidal, cycloidal, 5-D and 7-D polynomial trajectories and input-energies are compared in Fig. 5. In this case the car goes upward, and the dimensionless variables are $0 \leq \Theta \leq 1$ and $0 \leq \tau \leq 1$. We can find the dimensionless variables Θ_τ and $\Theta_{\tau\tau}$ of the 7-D polynomial has maximum values in Fig. 5(b) and (c), respectively. Since the initial external torque is negative and the string length decreases by the sheave rotation, the dimensionless variables

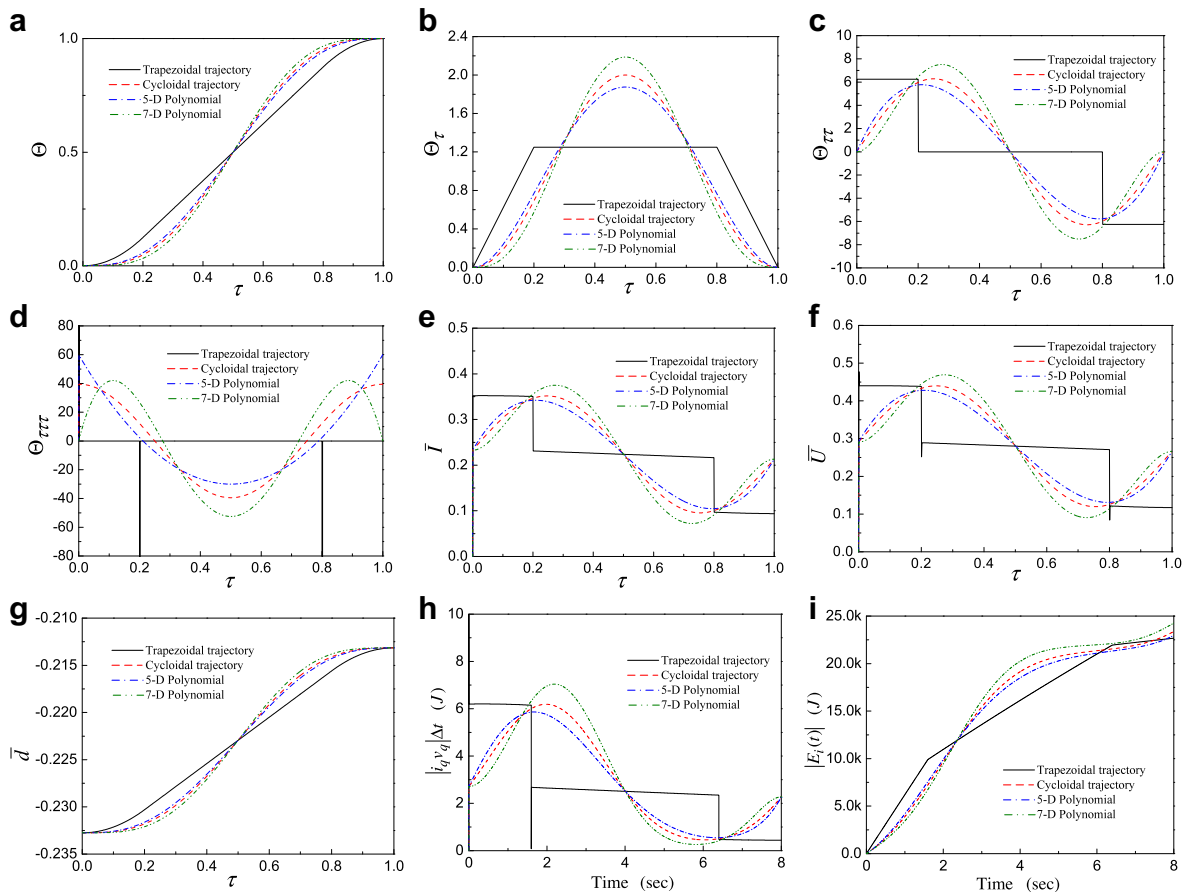


Fig. 5. The comparisons in the upward movement. (a) $\Theta - \tau$, (b) $\Theta_{\tau} - \tau$, (c) $\Theta_{\tau\tau} - \tau$, (d) $\Theta_{\tau\tau\tau} - \tau$, (e) $I - \tau$, (f) $\bar{U} - \tau$, (g) $\bar{d} - \tau$, (h) transient input energy, and (i) total absolute input energy.

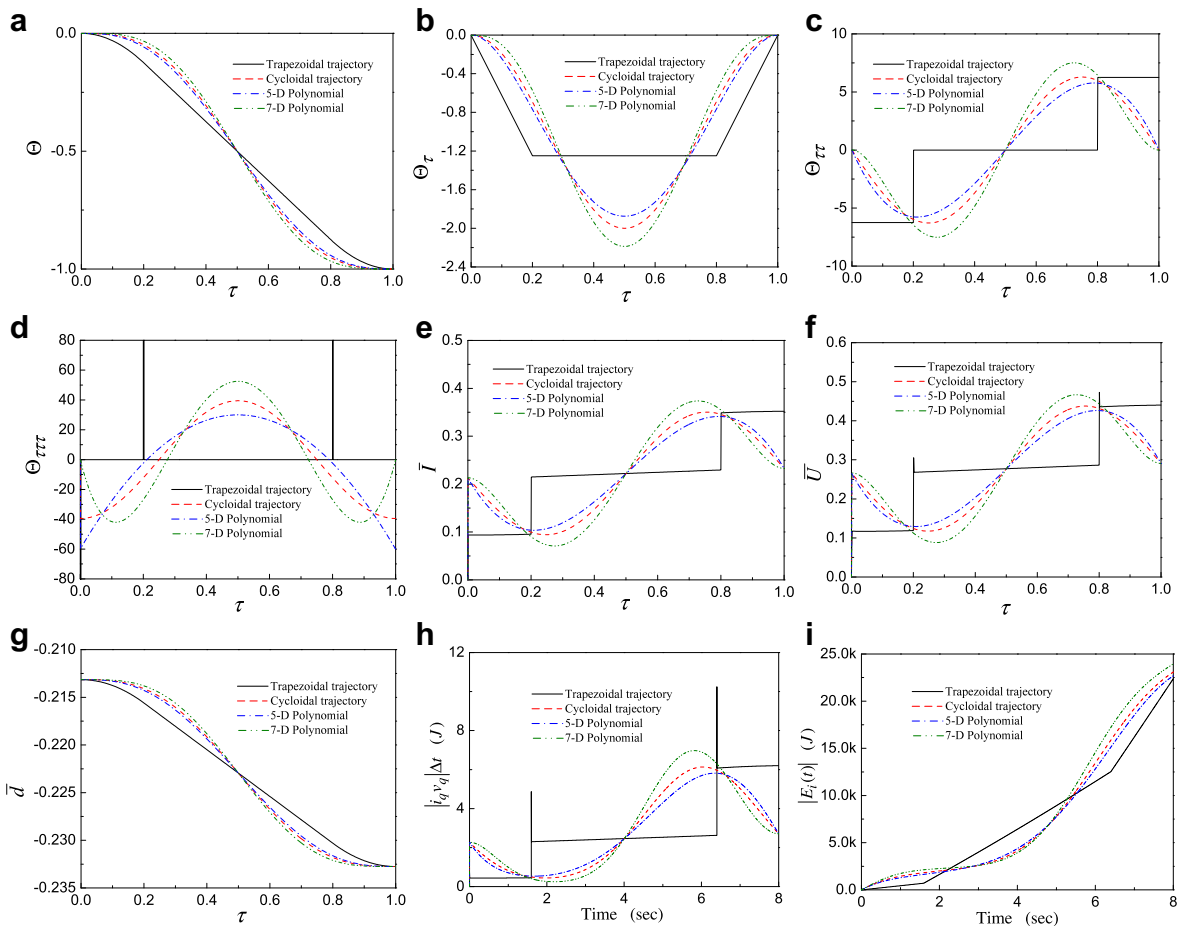
$\bar{I}$ and $\bar{U}$ in Fig. 5(e) and (f) have larger values before $\tau = 0.5$ than those after $\tau = 0.5$. Therefore, the dimensionless external torque $\bar{d}$ also decreases and is shown in Fig. 5(g). The dimensionless external torque $\bar{d}$ of the four trajectories is the same at the initial and final times, but is different and decreases from $\tau = 0$ to $\tau = 1$. It is found that the responses of the dimensionless external torque $\bar{d}$ are similar to the dimensionless variable Θ in Fig. 5(a), because the external torque is affected by the string length, and the string length is affected by the angular displacement. Finally, the transient absolute input energy and the total absolute input energy are shown in Fig. 5(h) and (i), respectively. The tendency of transient input energies of the cycloidal, 5-D and 7-D polynomial trajectories is similar to the dimensionless variables $\Theta_{\tau\tau}$, $\bar{I}$ and $\bar{U}$ in Fig. 5(c), (e) and (f). The reason is the differences of the external torques among the three trajectories are small; the angular acceleration is then the major affection factor for the input energy. From Fig. 5(i), it is found that the trapezoidal trajectory has minimum input energy, the 7-D polynomial has maximum one. The detail data are compared in Table 1. It can be found that the constraints in the initial and final states are less, the total absolute input energy are smaller. With the same constraints in initial and final states, the 5-D polynomial trajectory consumes less input energy than that of the cycloidal trajectory. It can be concluded from Fig. 5 and Table 1 that the relationship for the absolute input energy of the four trajectories from the first to third floor is the trapezoidal trajectory (minimum) < 5-D polynomial trajectory < cycloidal trajectory < 7-D polynomial trajectory (maximum).

In the contrary numerical simulations, the car goes downward, the departure floor is $n_i = 3$, arrival floor is $n_j = 1$ and the car average speed is $v_a = 1$ (m/s). The dimensionless transient responses are shown in Fig. 6, which is the reversed figure of Fig. 5. Since the car goes downward, the angular displacement and velocity are negative as shown in Fig. 6(a) and (b), respectively. The dimensionless variables Θ_{τ} and $\Theta_{\tau\tau}$ of the 7-D polynomial have maximum negative values in Fig. 6(b) and (c), respectively. The dimensionless variables I and $\bar{U}$ shown respectively in Fig. 6(e) and (f) for the four trajectories have small values before $\tau = 0.5$, but they have large values after $\tau = 0.5$. In the downward movement, the external torque is negative to the motor. From Eq. (20), we can find the dimensionless current variable I as shown in Fig. 6(e), and it produces a positive small brake torque for the motor initially and a positive large brake torque finally. In Fig. 6(g), it is found that the external torque is the major influence of the elevator system. Since the car goes downward, the negative external torque increases with the increasing length of the string. The tendency of the dimensionless variable $\bar{d}$ is also similar to the dimensionless

Table 1

The comparisons in the total absolute input energy and constraints (upward movement)

Trajectory	$ E_i $	Initial and final state constraints	Remark
Trapezoidal trajectory	22673.4175	4	Minimum
5-D Polynomial	22982.2533	6	
Cycloidal trajectory	23383.1347	6	
7-D Polynomial	24257.5341	8	Maximum

**Fig. 6.** The comparisons of the dimensionless variables with downward movement: (a) $\Theta - \tau$, (b) $\Theta_\tau - \tau$, (c) $\Theta_{\tau\tau} - \tau$, (d) $\Theta_{\tau\tau\tau} - \tau$, (e) $I - \tau$, (f) $\hat{V} - \tau$, (g) $\hat{D} - \tau$, (h) transient absolute input energy, and (i) total absolute input energy.

variable Θ in Fig. 6(a). Since the string mass is considered, the external torque is affected by the string length. Finally, the transient input energy and the total absolute input energy are compared in Fig. 6(h) and (i), respectively. The downward transient input energy before $\tau = 0.5$ in Fig. 6(h) is smaller than the upward transient input energy in Fig. 5(h). But after $\tau = 0.5$, the downward transient input energy in Fig. 6(h) is larger than the upward transient input energy in Fig. 5(h). It is found from Fig. 6(i) that the trapezoidal trajectory has minimum input energy, but the 7-D polynomial has maximum input energy. This result is the same with Fig. 5(i), and the detail data is compared in Table 2. Moreover, it is found from Tables 1 and 2 that the upward movement needs more input energy than that in the downward movement under the negative external torque.

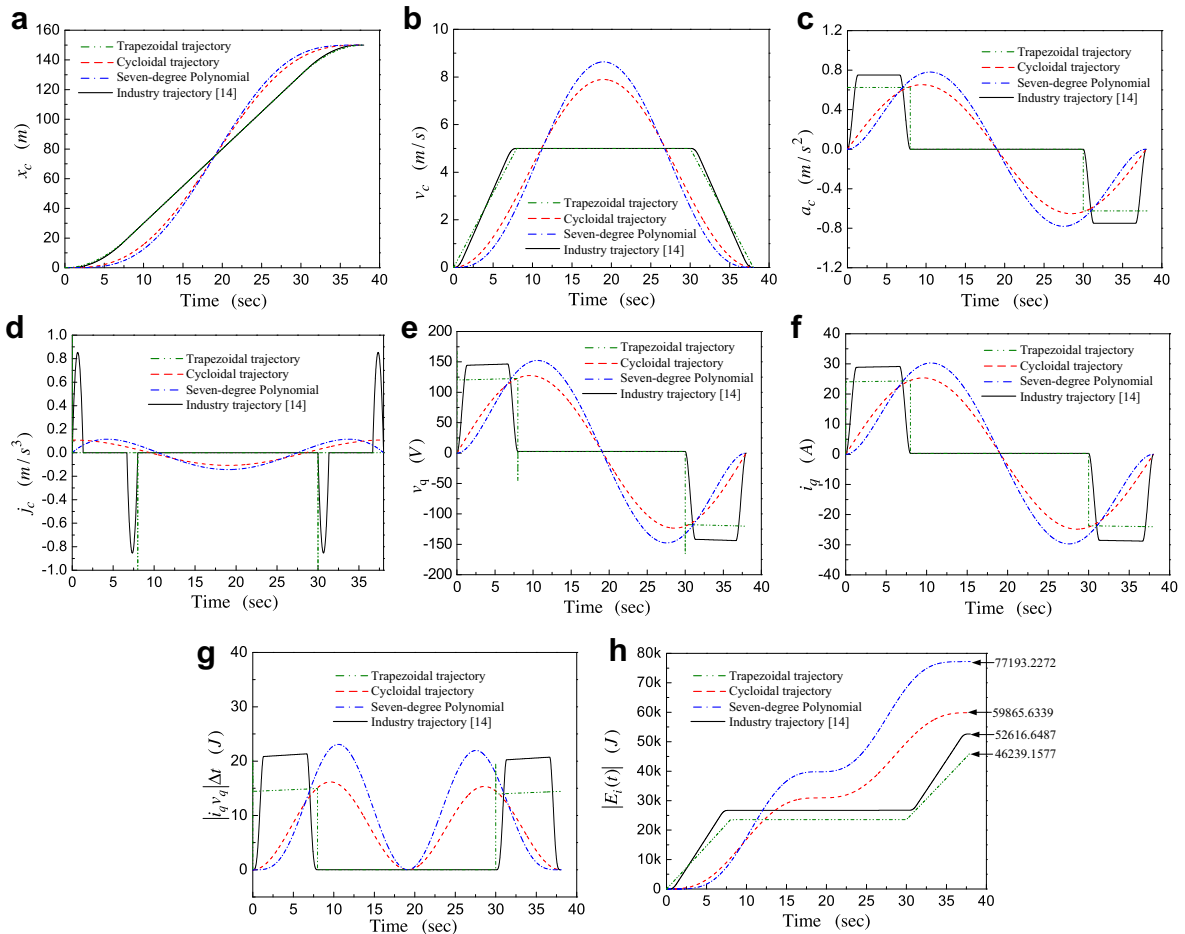
5.2. The comparison with industry trajectory

In the above cases, the four trajectories are compared in the upward and down movement. In this section, the trapezoidal, cycloidal and 7-D polynomial trajectories will be compared with the industry trajectory [14], which has the parameters:

Table 2

The comparisons in the total absolute input energy and constraints (downward movement)

Trajectory	$ E_t $	Initial and final state constraints	Remark
Trapezoidal trajectory	22381.0691	4	Minimum
5-D Polynomial	22684.7581	6	
Cycloidal trajectory	23085.6396	6	
7-D Polynomial	23960.0389	8	Maximum

**Fig. 7.** The comparisons of the four trajectories; (a) displacement, (b) speed, (c) acceleration, (d) jerk, (e) input voltage, (f) current, (g) transient input energy, and (h) total absolute input energy.

$h = 150$ (m), $T = 38$ (s), and $d(t) = 0$, and the initial and final constraint conditions: $x_c(0) = 0$ (m), $v_c(0) = 0$ (m/s), $a_c(0) = 0$ (m/s²), $j_c(0) = 0$ (m/s³), $x_c(38) = 150$ (m), $v_c(38) = 0$ (m/s), $a_c(38) = 0$ (m/s²) and $j_c(38) = 0$ (m/s³). The 7-D polynomial is chosen to compare, because it needs eight unknown coefficients, and can be solved exactly by the eight constraint conditions of the industry trajectory.

The car's responses of the four trajectories of displacement, speed, acceleration and jerk are compared in Fig. 7(a)–(d), respectively. The car displacement and speed in Fig. 7(a) and (b) respectively are almost the same for the industry and trapezoidal trajectories, where the trapezoidal trajectory has the following initial and final conditions: $x_c(0) = 0$ (m), $x_c(8) = 20$ (m), $x_c(30) = 110$ (m), $x_c(38) = 150$ (m), $v_c(0) = 0$ (m/s), $v_c(8) = 5$ (m/s), $v_c(30) = 5$ (m/s), and $v_c(38) = 0$ (m/s). These constraints are partially same with the industry trajectory, which has the following initial and final conditions: $a_c(0) = 0$ (m/s²), $a_c(8) = 0$ (m/s²), $a_c(30) = 0$ (m/s²), $a_c(38) = 0$ (m/s²), $j_c(0) = 0$ (m/s³), $j_c(8) = 0$ (m/s³), $j_c(30) = 0$ (m/s³) and $j_c(38) = 0$ (m/s³). These characteristics of constraint conditions can be found in Fig. 7(c) and (d). The input voltages, currents, transient and total input energies of the four trajectories are compared in Fig. 7(e)–(h), respectively. It is found that the tracks of the input voltages and currents are similar with the acceleration in Fig. 7(c). Therefore, the car acceleration is the major influence to the input

Table 3

The comparisons in the absolute input energy and constraints.

Trajectory	$ E_i $	Initial and final state constraints	Remark
Trapezoidal trajectory	46239.1577	4	Minimum
Reference trajectory [14]	52616.6487	8	
Cycloidal trajectory	59865.6339	6	
7-D Polynomial	77193.2272	8	Maximum

voltages and currents. The comparisons of the input energies are significant among the four trajectories, and are shown in Fig. 7(g) and (h). It is found that the transient input energies are near null for the trapezoidal and industry trajectories from 8 to 30 s, since the acceleration is null during these times. It can be explained that the car acceleration is the major energy consumption for the mechatronic system. The total input energy is shown in Fig. 7(h). It is seen that the trapezoidal trajectory consumes the minimum absolute input energy, but the 7-D polynomial consumes the maximum one. Table 3 describes the detailed comparison for the energies of the four trajectories. It is found that the trapezoidal trajectory has minimum absolute input energy because it has minimum constraint conditions; 7-D polynomial trajectory has maximum absolute input energy because it has maximum constraint conditions. Although the industry trajectory also has eight constraint conditions, it is restricted to the maximum speed and acceleration, and then it consumes less input energy than that of the 7-D polynomial trajectory.

5.3. SMC

In the SMC numerical simulations, we choose the departure floor $n_i = 1$, arrival floor $n_j = 3$ and the car average speed $v_a = 1$ (m/s). The dimensionless transient responses of the trapezoidal, 5-D and 7-D polynomial trajectories are the desired trajectories $\bar{\mathbf{X}}^*$. In this case the car goes upward, and the dimensionless variables are $0 \leq \Theta \leq 1$ and $0 \leq \tau \leq 1$. The comparisons of the SMC tracking control with $\Delta\bar{D} = 0$ ($\Delta m_c = 0$ kg) are shown in Fig. 8. It is found the tracking errors of Θ , Θ_τ and $\bar{I}$ converge to small ranges. The sliding surface s also converges to a small value. The control input $\bar{U}$ and total absolute input energy in Fig. 8(e) and (f) are similar to Fig. 5(f) and (i), respectively. It means that the proposed SMC can track the desired trajectories well and the trapezoidal trajectory also has the minimum absolute input energy.

The comparisons of the SMC tracking control with disturbance $\Delta m_c = 50$ kg are shown in Fig. 9. It is found that the proposed SMC also has the robustness and well tracking performances in Fig. 9(a)–(d). The control input $\bar{U}$ and total absolute input energy in Fig. 9(e) and (f) are larger than those in Fig. 8(e) and (f). It shows that the passenger mass Δm_c especially affects the control input and total absolute input energy. Under the loading condition ($\Delta m_c = 50$ kg), the trapezoidal trajectory tracked by the SMC also has the minimum absolute input energy and the 7-D polynomial has the maximum one.

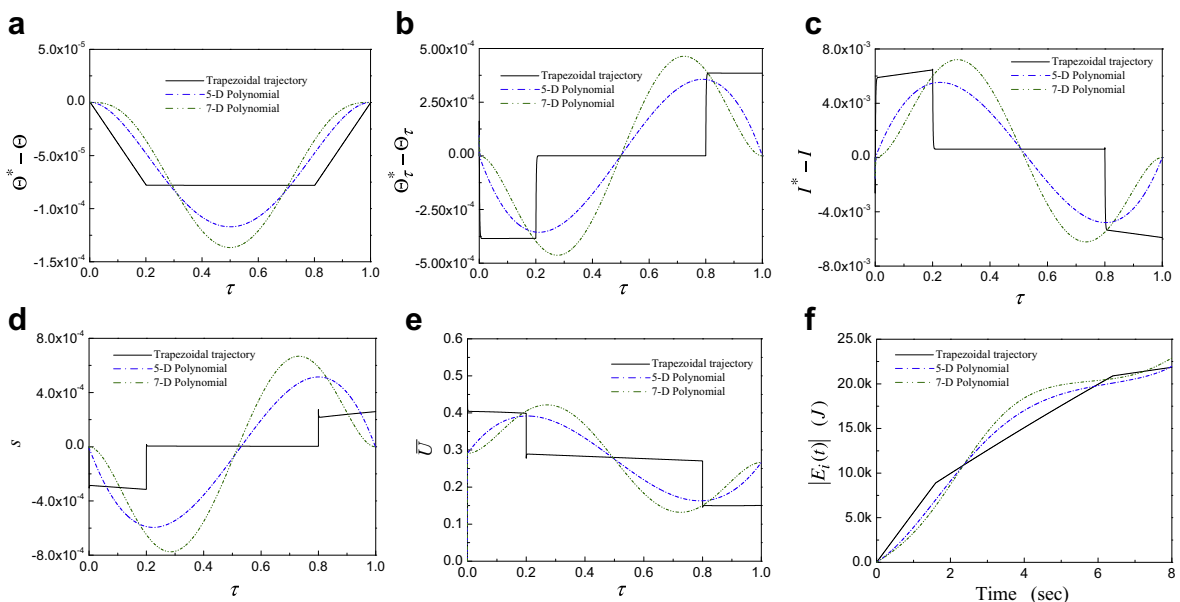


Fig. 8. The numerical comparisons by SMC among the trapezoidal, 5-D polynomial and 7-D polynomial trajectories with $\Delta\bar{D} = 0$ ($\Delta m_c = 0$ kg) in the upward movement; (a) errors of $\Theta - \tau$, (b) errors of $\Theta_\tau - \tau$, (c) errors of $I - \tau$, (d) sliding surface $s(\tau) - \tau$, (e) $\bar{U} - \tau$, (f) total absolute input energy.

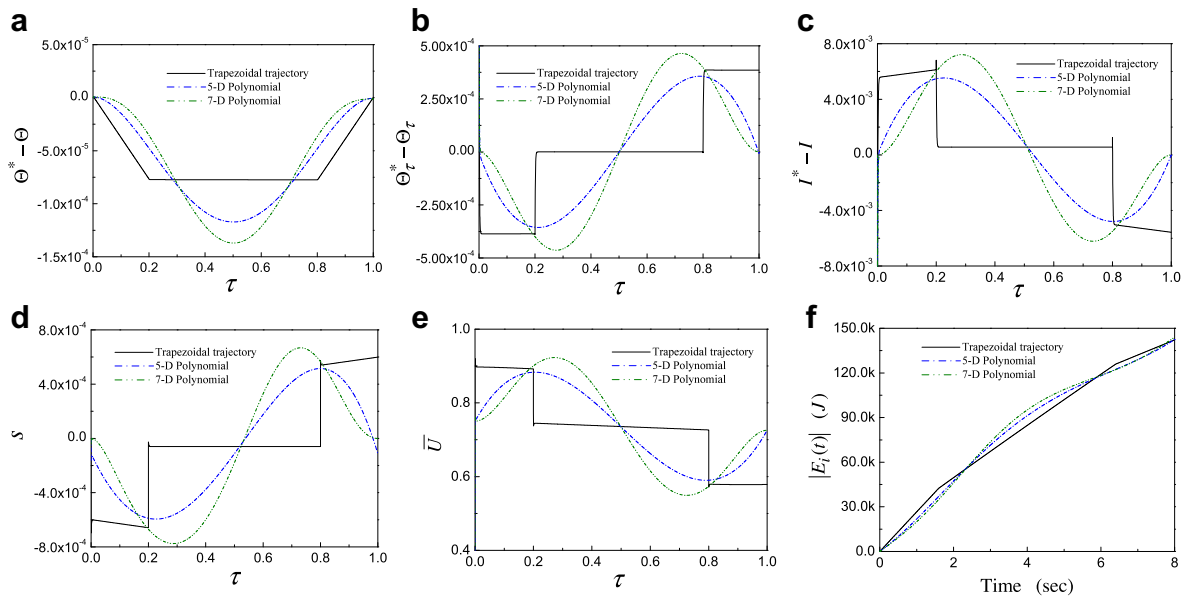


Fig. 9. The numerical comparisons by SMC among the trapezoidal, 5-D polynomial and 7-D polynomial trajectories with $\Delta\bar{D} = \Delta m_c g R / n K_{t i m} (\Delta m_c = 50 \text{ kg})$ in the upward movement; (a) errors of $\Theta - \tau$, (b) errors of $\dot{\Theta} - \dot{\tau}$, (c) errors of $\dot{l} - \dot{\tau}$, (d) sliding surface $s(\tau) - \tau$, (e) $\dot{U} - \tau$, (f) total absolute input energy.

5.4. Discussion and summary

From the comparisons between the car going upward and downward the first and third floors, the simulation results are described and summarized in Tables 1 and 2. Under the negative external torque, it can be found that the upward movement consumes more input energy than the downward movement. It is seen the external torque plays the significant influence to the elevator system. In addition, the output torque of the motor is used to be the brake torque during the downward movement. In the upward or downward movement, it is found that the trapezoidal trajectory with less constraint conditions has minimum input energy, but the 7-D polynomial trajectory has maximum one. Moreover, under the same constraint conditions it is found that the 5-D polynomial trajectory has less input energy than that of cycloidal trajectory.

From the comparisons with industry trajectory, the numerical simulations are organized and shown in Table 3. Why the minimum and maximum absolute input energies occur respectively in trapezoidal and 7-D polynomial trajectories for the elevator system? Firstly, it is observed that the constraint conditions are four, six and eight at the initial and final times for the corresponding trapezoidal, cycloidal and 7-D polynomial trajectories. The sequence of total absolute input energies consumed in the elevator system is that the trapezoidal trajectory is the minimum, the cycloidal trajectory is the middle and the 7-D polynomial trajectory is the maximum. According to these comparison and discussion, it is found that the less constraint is at the initial and final times, the less total absolute input energy will be consumed.

Furthermore, the industry trajectory has eight constraints, and is divided into seven regions, where the maximum acceleration and speed are restricted. The energy is mainly consumed in regions 1–3 and regions 5–7, but only small energy is consumed in region 4 because that the acceleration is null in this region. The 7-D polynomial trajectory has also eight constraints but no restriction in maximum acceleration and speed, the acceleration appears in all duration times. Therefore, the absolute input energy of the industry trajectory is less than that of the 7-D polynomial trajectory under the same eight constraints.

Summary of the numerical simulations, it is found four conclusions as follows:

- (1). The less constraint is restricted at the initial and final times; the less total absolute input energy will be consumed.
- (2). Under the same constraint, if the trajectory is designed with constant-speed regions and restriction in the maximum acceleration and speed, the consumed input energy will be less than that without constant-speed region and restriction of the designed trajectory.
- (3). In the viewpoint of the human comfortable, the 7-D polynomial trajectory is the comfortable trajectory in the initial and final regions, because they have the smallest acceleration and jerk. But the 7-D polynomial trajectory has maximum absolute input energy, because it has maximum constraint conditions. In the middle region, the industry trajectory and trapezoidal trajectory are the comfortable trajectories for human, because that the acceleration and jerk are null. Therefore, the trajectory has less acceleration and jerk, the human is more comfortable during the car's movement.
- (4). The proposed SMC based on dimensionless elevator system is realized and it has the robustness tracking control performance numerically.

6. Conclusions

The elevator system driven by a PMSM is dynamically modeled, and input-energy comparisons are made in this paper. The mechatronic model includes both the mechanical and electrical equations, and their dimensionless forms are developed for the upward and downward movement. The electrical energy of the elevator system is defined, and the total input energy equals the sum of dissipation energy, potential energy and output energy. The four trajectories including trapezoidal, cycloidal, 7-D polynomial and industry trajectories are compared in dynamic responses and energy consumption. It is found that under the same car's displacement and travelling time, the trapezoidal trajectory has the minimum total input energy, while the 7-D polynomial trajectory has the maximum one. Finally, the proposed SMC is realized to demonstrate that the controller has the robustness and well tracking control performance numerically.

Acknowledgment

The authors are grateful to the National Science Council for the support under Contract No. NSC- 98-2221-E-327 -010 -MY3

References

- [1] J.K. Kang, S.K. Sul, Vertical-vibration control of elevator using estimated car acceleration feedback compensation, *IEEE Trans. Ind. Electron.* 47 (1) (2000) 91–99.
- [2] R.F. Fung, J.H. Lin, C.M. Yao, Vibration analysis and suppression control of an elevator string actuated by a PM synchronous servo motor, *J. Sound Vib.* 206 (3) (1997) 399–423.
- [3] S.R. Venkatesh, Y.M. Cho, J. Kim, Robust control of vertical motions in ultra-high rise elevators, *Control Eng. Pract.* 10 (2002) 121–132.
- [4] C.S. Kim, K.S. Hong, M.K. Kim, Nonlinear robust control of a hydraulic elevator: experiment-based modeling and two-stage Lyapunov redesign, *Control Eng. Pract.* 13 (2005) 789–803.
- [5] D. Sha, V.B. Bajic, H. Yang, New model and sliding mode control of hydraulic elevator velocity tracking system, *Simul. Pract. Theory* 9 (2002) 365–385.
- [6] J.S. Yu, S.H. Kim, B.K. Lee, C.Y. Won, J. Hur, Fuzzy-logic-based vector control scheme for permanent-magnet synchronous motors in elevator drive applications, *IEEE Trans. Ind. Electron.* 54 (4) (2007) 2190–2200.
- [7] M. Schlemmer, S.K. Agrawal, A computational approach for time-optimal planning of high-rise elevators, *IEEE Trans. Control Syst Technol.* 10 (1) (2002) 105–111.
- [8] N. Mutoh, N. Ohnuma, A. Omiya, M. Konya, A motor driving controller suitable for elevators, *IEEE Trans. Power Electron.* 13 (6) (1998) 1123–1134.
- [9] E. Polar, M. Deparis, An algorithm for minimum energy control, *IEEE Trans. Autom. Control* 14 (4) (1969) 367–377.
- [10] Y.E. Sahinkata, R. Asami, Minimum-energy control of a class of: electrically driven vehicles, *IEEE Trans. Autom. Control* 17 (1972) 1–6.
- [11] C. Canudas de Wit, S.I. Seleme Jr., Robust torque control design for induction motors: the minimum energy approach, *Automatica* 33 (1) (1997) 63–79.
- [12] P. Kokotovic, G. Singh, Minimum-energy control of traction motor, *IEEE Trans. Autom. Control* (1972) 92–95. Short Papers.
- [13] M.S. Huang, Y.L. Hsu, R.F. Fung, Minimum-energy point-to-point trajectory planning for a motor-toggle servomechanism, *IEEE/ASME Trans. Mechatron.* 17 (2) (2012) 337–344.
- [14] W.D. Zhua, L.J. Teppo, Design and analysis of a scaled model of a high-rise, high-speed elevator, *J. Sound Vib.* 264 (2003) 707–731.
- [15] H. Liu, X. Lai, W. Wu, Time-optimal and jerk-continuous trajectory planning for robot manipulator with kinematic constraints, *Rob. Comput. Integr. Manuf.* 29 (2013) 309–317.
- [16] A. Gasparetto, A. Lanzutti, R. Vidoni, V. Zanutto, Experimental validation and comparative analysis of optimal time-jerk algorithms for trajectory planning, *Rob. Comput. Integr. Manuf.* 28 (2012) 164–181.
- [17] A. Gasparetto, V. Zanutto, Optimal trajectory planning for industrial robots, *Adv. Eng. Softw.* 41 (2010) 548–556.
- [18] A. Gasparetto, V. Zanutto, A technique for time-jerk optimal planning of robot trajectories, *Rob. Comput. Integr. Manuf.* 24 (2008) 415–426.
- [19] A. Gasparetto, V. Zanutto, A new method for smooth trajectory planning of robot manipulators, *Mech. Mach. Theory* 42 (2007) 455–471.
- [20] M. Boryga, A. Grabos, Planning of manipulator motion trajectory with higher-degree polynomial use, *Mech. Mach. Theory* 44 (2009) 1400–1419.
- [21] R. Krishnan, *Electric Motor Drives: Modeling, Analysis and Control*, Prentice Hall, New York, 2001.
- [22] W.P. Graebel, *Engineering Fluid Mechanics*, Taylor and Francis, New York, 2001.
- [23] L. Biagiotti, C. Melchiorri, *Trajectory Planning for Automatic Machines and Robots*, Springer-Verlag, Berlin, Germany, 2008.